Collaborative Relation Augmentation With Hierarchical Prescription Inference for Medication Recommendation

Xiaobo Li, Xiaodi Hou, Fanjun Meng, Hai Cui, and Yijia Zhang

***Abstract*—Medication recommendation systems have emerged as crucial tools in healthcare, offering personalized and effective drug combinations tailored to individual patient's clinical profiles. However, most existing approaches primarily focus on drug prediction by analyzing patient-drug interactions, often neglecting the intricate correlations between diseases and drugs. To address above limitation, this paper proposes a novel Collaborative Relation augmentation with Hierarchical Prescription inference network (CRHP) for effective medication recommendation. CRHP first constructs multiple covariance knowledge graphs to capture fine-grained interaction relationships between different entities from a global perspective. Based on self-built knowledge graphs, CRHP designs a collaborative relation augmented learning module, which introduces hypergraph convolutional networks to capture high-order association information between different entities. Moreover, CRHP devises a hierarchical prescription inference module that formulates drug prescriptions based on both current and historical patient information. The extensive experiments on two publicly available real-world medical datasets, MIMIC-III and MIMIC-IV, demonstrate the effectiveness of CRHP. The results indicate significant performance improvements over baseline methods, with gains of 2.12 and 1.31 in Jaccard, 1.91 and 1.83 in PRAUC, and 1.79 and 0.98 in F1-score (in percentage points).**

***Index Terms*—Medication recommendation, healthcare, hypergraph convolutional network.**

I. Introduction

ADVANCEMENTS in technology and digitization, particularly in artificial intelligence and deep learning, have enabled the efficient processing of large-scale Electronic Health Records (EHRs), thereby supporting healthcare analytics and decision-making [1]. Extensive EHR databases encompass detailed information about patient visits across various time points, medical procedures, laboratory tests, disease diagnoses, prescribed medications, and other pertinent details [2]. Research endeavours concerning EHRs have diversified, encompassing tasks such as disease diagnosis [3], outcome prediction [4], medical code prediction [5], and radiology report generation [6], [7]. Medication recommendation systems play a pivotal role in aiding physicians in formulating accurate and efficient prescribing decisions [8].

Medication recommendation aims to offer a personalized combination of drugs for treating a patient's diagnosed disease (e.g., diagnosis and procedure codes) [9]. Early works [10], [11] on medication recommendation overlook the sequential nature of the problem by solely focusing on the current patient visit. These methods fail to mine the valuable information in historical visits, which could be leveraged to improve the personalization of medication recommendations. Subsequent researches [12], [13] integrate patient longitudinal historical information to comprehensively model patient health status, obtaining more effective recommendation outcomes. In addition, medical knowledge, such as drug molecular structure [14], drug dosage titrations [15] and symptom data [16], has also been widely introduced to enhance drug representation learning.

Existing drug recommendation systems, despite significant progress, still face challenges in effectively modeling complex disease-drug correlation patterns. Specifically, prevalent approaches [17], [18] typically involve a two-step process: initially, diagnosis and procedure sequences are individually represented, and then these feature representations are concatenated to describe the patient's health status. Subsequently, medication predictions are formulated through the interaction between the patient and drug representations. Consequently, previous methods primarily focus on enhancing the quality of patient and drug representations to improve drug prediction accuracy. However, these methods often fail to explore the correlations between diseases and drugs, which remains to be utilized within the prescription formulation to improve the feature representation.

This paper proposes a novel Collaborative Relation augmentation with Hierarchical Prescription inference network (CRHP) to produce the effective medication recommendation. Specifically, CRHP first constructs multiple Covariance Knowledge Graphs (CKGs), i.e. diagnosis-medication covariance knowledge graph (dm-CKG) and procedure-medication covariance knowledge graph (pm-CKG) to integrate knowledge relationships between

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Xiaobo Li, Hai Cui, and Yijia Zhang are with the School of Information Science and Technology, Dalian Maritime University, Dalian 116026, China (e-mail: xiaobo.li@dlmu.edu.cn; zhangyijia@dlmu.edu.cn; cuihai@dlmu.edu.cn).

Xiaodi Hou and Fanjun Meng are with the School of Artificial Intelligence, Dalian Maritime University, Dalian 116026, China (e-mail: houxiaodi@dlmu.edu.cn; m1120220648@dlmu.edu.cn).

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different entities from a global perspective. Based on self-built CKGs, CRHP designs a collaborative relation augmented module and utilizes hypergraph convolutional networks to capture high-order association information between multiple entities. Additionally, to enrich drug representation from different perspectives, CRHP designs a hierarchical prescription inference module that predicts drugs based on current and historical medical information. Finally, considerable experiments on real-world datasets validate the proposed model's efficacy.

In summary, the main contributions of the model CRHP are as follows:

- This paper first constructs multiple covariance knowledge graphs that incorporate fine-grained and global interaction relationships among diverse entities to enhance feature representation.
- This paper proposes a collaborative relation augmented learning module, which utilizes hypergraph convolutional networks to capture high-order association information.
- This paper designs a hierarchical prescription inference module that employs a dual-view prediction strategy to comprehensively model drug representation from current and past medical information.
- When evaluated on benchmark dataset, i.e. MIMIC-III and MIMIC-IV datasets, the proposed CRHP achieves new state-of-the-art performance in major metrics.

The remainder of this paper is organized as follows. Section II reviews the related work, providing a comprehensive overview of existing research. Section III first defines the structure of EHR data, then details the knowledge graph construction process, and finally formulates the medication recommendation problem. Section IV presents the proposed methodology. Section V describes the experiments, including the experimental setups and comparative analysis of results. Finally, Section VI concludes the paper by summarizing the key findings and contributions, while Section VII discusses the limitations of the proposed approach and outlines potential directions for future research.

II. Related Work

A. Medication Recommendation

Most existing drug recommendation methods typically utilize medical information from EHR for feature representation, capturing potential interaction mechanisms between patients and drugs. These methods mainly include instance-based recommendation methods and longitudinal-based recommendation methods.

The instance-based approaches [10], [19], [20] focus on current visit information and fail to explore patients' historical medical records. For example, Leap [10] first encodes the patient's current visit information into a structured representation before decomposing treatment recommendations through a sequential decision process. The framework employs a recurrent decoder with content-based attention to model medication dependencies while introducing reinforcement learning guided by clinical knowledge to optimize precision and drug safety. CompNet [20] addresses order-free medicine combination prediction through a reinforced graph architecture, dynamically fusing patient-specific EHR features with knowledge graph embeddings while enforcing Drug-Drug Interaction (DDI) safety. However, the instance-based drug recommendation method only focuses on the medical representation of the current visit, neglecting the modeling of historical information of patients who have visited multiple times.

On the contrary, longitudinal methods [21], [22], [23], [24] aim to capture long-term dependencies across multiple clinical visits, providing additional valuable information supplements for patients' current visits to develop personalized medication by mining the historical information. For example, MGRN [22] uses RNNs to model the dynamic dependencies between diagnosis and procedure during multiple visits, constructing a comprehensive patient representation. VITA [23] proposes a drug recommendation framework based on relevant visit selection and a target-aware attention mechanism. This framework models the patient's health status by dynamically evaluating the correlation between current and historical visits, improving recommendation accuracy.

In addition, multiple models [14], [25], [17], [26], [27] integrate medical knowledge and utilize graph representation learning techniques to enhance feature representation and improve drug recommendation performance. For instance, GAMENet [17] designs a graph-augmented memory network to incorporate an EHR knowledge graph and a DDI graph, offering safe and effective medication recommendations. SafeDrug [14] and MoleRec [25] integrate drug molecular substructures to learn the relevancy between the patient's health status and substructures and achieve improved performance. DAI-Net [26] introduces implicit symptom knowledge and constructs a dual adaptive interaction network. This network extracts the association between symptoms and drug molecules through a symptom-aware drug-matching module to improve the model's performance and robustness. However, these approaches rarely explicitly model the associations between diseases and drugs, leaving this aspect unexploited in optimizing feature representation for prescriptions.

B. Graph Neural Network

The convolutional Graph Neural Network (GNN) has garnered considerable attention across diverse domains. By aggregating features from neighbouring nodes across the graph, the convolutional GNN can learn robust node representations. Recently, several studies [2], [12], [17], [28] have explored the utilization of GNNs to enhance medication recommendation systems. For instance, GAMENet [17] utilizes the Graph Convolutional Network (GCN) [29] to encode DDI knowledge and drug co-occurrence knowledge to improve medication representation. PREMIER [2] employs two graph attention networks to encode DDI and EHR knowledge, learning an efficient drug representation. SafeDrug [14] encodes the molecular structure of drugs by employing a message-passing neural network to enhance drug recommendation safety. However, these methodologies primarily focus on drug interactions as nodes for information aggregation, overlooking complex relationships among other medical entities. Unlike conventional graph structures,

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hypergraphs offer greater flexibility by allowing hyperedges to connect multiple nodes simultaneously, thus effectively modeling higher-order dependencies among diverse entities [30], [31]. Building on this advantage, CRHP leverages a hypergraph convolutional network to comprehensively represent multi-entity relationships, while integrating knowledge graph information to enhance drug recommendation accuracy.

III. Problem Formulation

Electrical Health Records (EHR): An EHR system, represented as $\mathcal{R} = \{V^{(t)}\}_{k=1}^{N_k}$, contains a set of medical records for N_k patients. In this context of the drug recommendation, a patient may consist of several visits $V = \{V^1, V^2 \cdots, V^T\}$, where T represents the total visit times for k-th patient. The patient's t-th visit can be expressed as $V^t = \{D^t, P^t, M^t\}$, where $D^t \in \{0,1\}^{|D|}$, $P^t \in \{0,1\}^{|P|}$ and $M^t \in \{0,1\}^{|M|}$ are the set of diagnosis, procedure and medication, where a value of 1 means that the medication is prescribed, while 0 denotes the medication's absence, and $|D|$, $|P|$ and $|M|$ are the number of the diagnosis, procedure and medication in each sequence.

Knowledge Graph Construction: Let $G_{dm} = (h_d, r_d, t_d)$ and $G_{pm} = (h_p, r_p, t_p)$ denote the covariance knowledge graphs (dm-CKG and pm-CKG), which organize different types of entities (i.e. diagnosis, procedure and medication) and corresponding relationships. Specifically, each entity-relation-entity triplet (h, r, t) characterizes the relatedness between the head entity h and tail entity t with the relation r. Such information incorporates disease and connections among drugs as side information to enhance the modeling of disease-drug interaction for recommendations. The specific construction process is as follows.

Given a dataset comprising N visits, each diagnosis entity d_i during the n-th visit is represented by a binary variable d_i^n. Specifically, $d_i^n = 1$ if the diagnosis entity appears in the n-th visit, and $d_i^n = 0$ otherwise. An analogous definition applies to procedure and medication entities. In statistics, the correlation coefficient is derived from the covariance between nodes. The covariance $Cov(d_i, m_j)$ between the i-th diagnosis entity d_i and the j-th medication entity m_j can be expressed as:

$$\bar{d}_i = \frac{\sum_{n=1}^{N} d_i^n}{N}; \bar{m}_j = \frac{\sum_{n=1}^{N} m_j^n}{N}, \tag{1}$$

$$Cov(d_i, m_j) = \frac{\sum_{n=1}^{N} (d_i^n - \bar{d}_i)(m_j^n - \bar{m}_j)}{N-1}, \tag{2}$$

where $\bar{d}_i$ represents the mean occurrences of d_i and m_j across all visits, respectively. The above process yields the correlation coefficient matrix $A^{(d,m)}$, as well as the correlation coefficient matrix $A^{(p,m)}$ between all procedure entities and medication entities.

Medication Recommendation Problem: Existing models [32], [33] are mainly based on the patient's diagnosis and procedure ICD (International Classification of Diseases) codes to generate medication recommendations. The model input includes the diagnosis sequence D^t and procedure sequence P^t of the patient's current visit V^t and the diagnosis, procedure, and medication sequences of previous visits $\{V^1, V^2, \ldots, V^{t-1}\}$. Specifically, the ICD code sequence of diagnosis and procedure D^t and P^t are first projected into an embedding space and then concatenated to form the patient representation, which is then matched with medication representation M_e to obtain the final predicted medication combination $\hat{M}^t$.

IV. Methodology

A. Overall Architecture

This section provides a detailed introduction to the proposed model, CRHP, which is mainly composed of two key modules: collaborative relation augmented learning module and hierarchical prescription inference module. Fig. 1 provides a schematic overview of the model, while the following section details each constituent part.

B. Collaborative Relation Augmented Learning

In this part, CRHP designs a Collaborative Relation Augmented Learning module (CRAL) to address such limitations, as previous approaches encode disease and drug entities that rely on random initialization and embed these entities independently, disregarding the intrinsic interrelations among different entities that such random embeddings fail to capture. Therefore, CRAL aims to learn local and global collaborative interaction relations between diseases and drugs based on the self-constructed medical knowledge graphs. Additionally, CRAL effectively integrates the hypergraph structure encoding to comprehensively capture the complex high-order dependencies among different medical entities, reinforcing the representation quality of drug recommendation. This module consists of three components: local relation encoding, hypergraph global dependency and multi-order aggregation. The whole process is divided into two branches: diagnosis-medication and procedure-medication, which respectively explore the potential relationships between the two types of entity pairs. For simplicity, the following description focuses specifically on the diagnosis-medication branch.

1) Local Relation Encoding: To incorporate the direct interactions between medical entities from self-constructed medical knowledge graphs, CRAL designs a Local Relation Encoding component (LRE) based on graph propagation. Unlike traditional methods [17], [18] that initialize embeddings randomly or treat entities independently, the proposed model CRHP explicitly captures entity dependencies through a lightweight graph convolutional network.

Specifically, LRE uses $E^{(d)}$ and $E^{(m)}$ as the embedding matrices for diagnosis and medication entities, where the i-th row vector $e_i^{(d)}$ in $E^{(d)}$ and the j-th row vector $e_j^{(m)}$ in $E^{(m)}$ are defined as the embedding representations of the i-th diagnosis entity and the j-th medication entity, respectively. Drawn inspiration from the proven efficacy of the simplified GNN employed in LightGCN [34], LRE uses this approach to represent the embeddings associated with diagnoses and medications. The local graph embedding layer is devised in the following manner:

$$Z^{(d)} = \sigma_1(\bar{A}^{(d,m)} \cdot E^{(d)}); Z^{(m)} = \sigma_1((\bar{A}^{(d,m)}) \cdot E^{(m)}), \tag{3}$$

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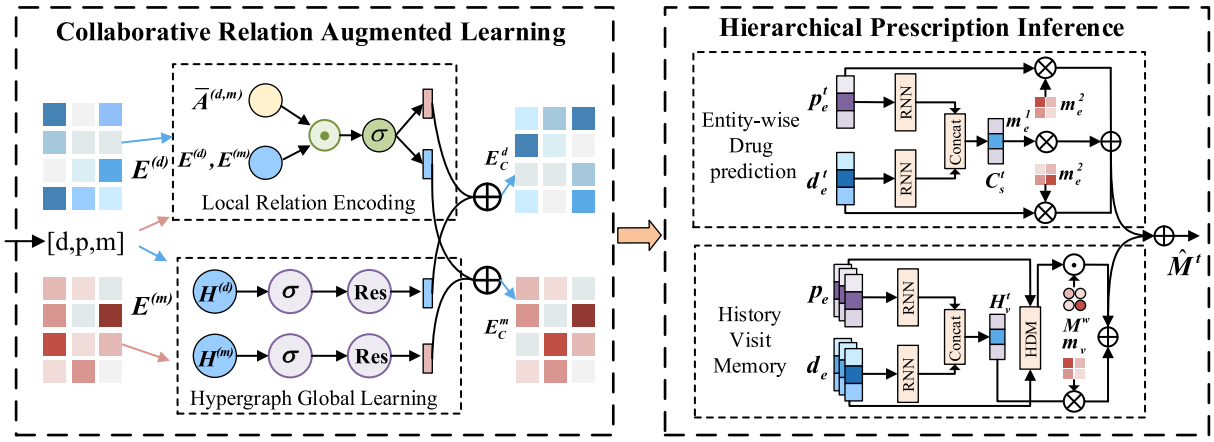


Fig. 1. The schematic illustration of the proposed CRHP. Based on the customized covariance knowledge graph, the collaborative relation augmented learning is proposed to learn medical entity embeddings via mining relations among various nodes. Finally, the hierarchical prescription inference schema is devised to comprehensively model drug representation from current and past medical information.

where $Z^{(d)}$ and $Z^{(m)}$ represent the aggregated representations of diagnosis and medication entities respectively in the diagnosis-medication branch, $\sigma_1(\cdot)$ denotes the LeakyReLU activation function with a slope for negative inputs, which benefits gradient backpropagation and introduces non-linearity into the transformation. Notably, $\bar{A}^{(d,m)}$ is obtained from the correlation coefficient matrix $A^{(d,m)}$, as follows:

$$\bar{A}^{(d,m)} = D_{(d)}^{-1/2} \cdot A^{(d,m)} \cdot D_{(m)}^{-1/2}, \tag{4}$$

where $D_{(d)}^{-1/2}$ and $D_{(m)}^{-1/2}$ are the diagonal degree matrices corresponding to the diagnosis and medication entities, respectively.

Subsequently, LRE utilizes multiple embedding propagation layers to aggregate local neighborhood information for the contextual embedding representation generation. Moreover, LRE leverages residual connections to integrate self-information during message aggregation across graph layers, emphasizing central node semantics and mitigating GNN's over-smoothing. The message passing from $(l-1)$-th to the (l)-th layer is denoted as follows:

$$E_l^{(d)} = Z_l^{(d)} + E_{l-1}^{(d)}; E_l^{(m)} = Z_l^{(m)} + E_{l-1}^{(m)}. \tag{5}$$

2) Hypergraph Global Learning: Motivated by the capacity of hypergraphs to unify nodes beyond pairwise relationships [35], CRHP incorporates a deep Hypergraph Global Learning (HGL) component to capture intricate high-order relationships. A hypergraph comprises a set of vertices and hyperedges, where each hyperedge has the flexibility to connect an arbitrary number of vertices [36]. Therefore, HGL employs hyperedges as intermediary centres to facilitate global-aware information dissemination between diagnoses and drugs, transcending the limitation of hop distance.

Specifically, HGL defines the hypergraph dependency matrices $H^{(d)} \in \mathbb{R}^{D \times h_d}$ and $H^{(m)} \in \mathbb{R}^{M \times h_m}$ for diagnoses and drugs, respectively, where h_d and h_m denote the number of hyperedges. HGL views hypergraph dependency matrices as $H^{(*)}$ and presents a formal formulation of the hypergraph message passing mechanism as follows:

$$\tilde{\Lambda} = \phi^c(\Lambda), \varphi(X) = \sigma_1(WX) + X, \Lambda = H^T \cdot E, \tag{6}$$

$$\Upsilon_l = \sigma_1(H^{(*)} \cdot \tilde{\Lambda}), \tag{7}$$

$$\Upsilon_l^{(d)} = \sigma_1(H^{(d)}\Lambda^{(d)}); \Upsilon_l^{(m)} = \sigma_1(H^{(m)}\Lambda^{(m)}), \tag{8}$$

where $\Lambda^{(d)}$ and $\Lambda^{(m)}$ represent the hyperedge-specific embeddings for diagnoses and medications, respectively. $\Upsilon_l^{(d)}$ and $\Upsilon_l^{(m)}$ denote the l-th propagation layer of hypergraph embeddings about diagnoses and drugs in hypergraph representation space, respectively.

3) Multi-Order Aggregation: Finally, the model integrates local relation encoding with global dependency modeling through iterative execution of graph-based local embedding propagation and hypergraph information aggregation, as follows:

$$E_l^d = Z_l^{(d)} + \Upsilon_l^{(d)} + E_{l-1}^{(d)}; E_l^m = Z_l^{(m)} + \Upsilon_l^{(m)} + E_{l-1}^{(m)}, \tag{9}$$

where $E_l^{(d)}$ denotes the l-th layer embedding of diagnosis entities, $E_l^{(m)}$ is the l-th layer embedding of medication entities.

Furthermore, residual operations are further applied during the embedding aggregation, as follows:

$$E_C^d = \sum_{l=1}^{L} E_l^d; E_C^m = \sum_{l=1}^{L} E_l^m, \tag{10}$$

where E_C^d represents the final diagnostic entity embedding containing the diagnosis-medication dependency relationship. Similarly, by performing the above calculation steps, CRHP can obtain procedure entity embeddings E_C^p that contain procedure-medication dependency relationships.

C. Hierarchical Prescription Inference

To improve drug prediction performance, CRHP designs a Hierarchical Prescription Inference (HPI) module to formulate drug prescriptions based on both current and historical patient

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information. HPI first constructs a patient visit embedding representation based on the medical entity embedding representation obtained from CRAL. Subsequently, entity-wise drug prediction and history visit memory are designed to predict drugs based on current and historical medical information, respectively.

1) Medical Entity Embedding: As mentioned before, a patient's single visit V^t, which consists of three multi-hot vector sequences, denoted as $V^t = [d^t, p^t, m^t]$. The diagnosis and procedure entities are mapped into embedding representation by two vocabularies E_C^d, E_C^p, as follows:

$$\begin{cases} d_e^t = Emb_d(d^t, E_C^d), \\ p_e^t = Emb_p(p^t, E_C^p), \end{cases} \tag{11}$$

where Emb_d and Emb_p are the embedding operation that can project medical entity into the embedding space, $d_e^t \in \mathbb{R}^{S_d \times dim}$ and $p_e^t \in \mathbb{R}^{S_p \times dim}$, S_d and S_p represent the number of medical entities in d^t and p^t, respectively.

2) Entity-Wise Drug Prediction: To capture the interaction between different entities and drug representations, HPI designs a two-stage training strategy. In stage-1, HPI aims to model the long-term dependency relationship between diagnosis and procedure entity sequences, integrating entity-wise interactions within the disease sequence for drug prediction. The stage-2 aims to directly capture entity-drug associations, alleviating unnecessary information loss in previous sequence learning during network propagation.

In stage-1, HPI first utilizes Recurrent Neural Networks (RNNs) to extract long-range dependency between multiple entities in diagnosis and procedure entity sequences. Then, HPI concats diagnosis and procedure sequence representation to obtain the current patient representation. The calculation process can be represented as:

$$d_s^t = \text{RNN}_1(d_e^t); p_s^t = \text{RNN}_2(p_e^t), \tag{12}$$

$$C_s^t = Concat(d_s^t, p_s^t), \tag{13}$$

where $Concat(\cdot)$ represents the concat operation and $C_s^t \in \mathbb{R}^{2dim}$ denotes the patient representation.

Motivated by previous work [22], HPI introduces randomly initialized drug representations m_e^1 via standard normal distribution, which are progressively refined through gradient-based optimization during training. These embeddings dynamically adapt through interactions with patient states C_s^t to generate context-aware drug matching score m_s^t, as follows:

$$m_s^t = C_s^t m_e^1, \tag{14}$$

where the drug-matching score serves as intermediate predictive signals that bridge patient-specific representations with final medication recommendations.

Unlike the first stage, the second stage aims to connect individual medical elements within the relevant input sequence with drug representations earlier instead of creating a broad representation of the patient's disease sequence for overall interaction. This approach endeavours to mitigate the loss of disease-drug interaction information during the iterative propagation through deep networks, learning more detailed and precise drug representations tailored to specific diseases.

Therefore, HPI directly links each diagnosis and procedure to the drug representation m_e^2 at the entity level. This ensures that the unique meaning of each entity is preserved as much as possible. The calculation process is as follows:

$$d_t^t = d_e^t m_e^2; p_t^t = p_e^t m_e^2. \tag{15}$$

Finally, HPI combines the output of two stages and gets the module's drug-matching score $E_m^t \in \mathbb{R}^{|M|}$:

$$E_m^t = MLP(d_t^t + p_t^t) + m_s^t, \tag{16}$$

where $MLP(\cdot)$ denotes Multi-Layer Perceptron, which is used to adjust model's focus on features.

3) History Visit Memory: Predicting drug scores based on current medical entity information is insufficient and neglects the integration of patient historical information. Therefore, HPI designs a History Visit Memory (HVM) component to extract valuable information from the patients' historical disease record perspective and historical drug record perspective to obtain information-rich drug representations.

Disease Perspective: First, HPI leverages RNN to encode diagnosis and procedure sequences across multiple visits, capturing long-range dependency among visits. Then, HPI concats the encoded representations to acquire comprehensive patient representation. The calculation process is as follows:

$$\begin{cases} (D_v^1, \ldots, D_v^t) = \text{RNN}_{vd}((d_v^1, \ldots, D_v^t)), \\ (P_v^1, \ldots, P_v^t) = \text{RNN}_{vp}((P_v^1, \ldots, P_v^t)), \end{cases} \tag{17}$$

$$H_v^t = Concat(D_v^t, P_v^t), \tag{18}$$

where H_v^t denotes the patient representation integrated history disease record.

Next, HPI acquires the drug-matching score for the disease-level perspective, as follows:

$$r_h^t = H_v^t m_v. \tag{19}$$

Medication Perspective: In this perspective, HPI aims to extract essential historical drug information across multiple visits, thereby facilitating the recommendation of more rational drug combinations for the patient's current visit. Initially, HPI concats the data from the current visit with each of the historical visits to create a comprehensive History Drug Memory (HDM). Following this, HPI incorporates a readout function $Readout(\cdot)$ to discern informative gating memory representations.

$$\alpha^w = Readout(Concat(C^t, C^w)), \tag{20}$$

where $\alpha^w \in \mathbb{R}^{(t-1) \times N_m}$ represents historical medication score weight, $w \in [1, t-1]$.

Next, HPI computes the dot product between the historical memory weights and the historical drug sets to derive the scores for the recommended drugs, denoted as r_m^t, as follows:

$$r_m^t = \sum_{w=1}^{t-1} \alpha^w \odot M^w, \tag{21}$$

where M^w denotes the drug set at the w-th history visit.

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D. Drug Representation Aggregation

Finally, HPI aggregates all the drug-matching score vectors as the global drug-match probability:

$$\hat{M}^t = \sigma_2(E_m^t + r_h^t + r_m^t), \tag{22}$$

where σ_2 is Sigmoid activation function and $\hat{M}^t$ denotes the final medication prediction set.

E. Training Objective

The medication recommendation task is formulated as a multi-label prediction problem, with model convergence constrained through a combined loss function incorporating binary cross entropy loss $\mathcal{L}_{bce}$, multi-label hinge loss $\mathcal{L}_{mlh}$, and DDI loss $\mathcal{L}_{ddi}$ to constrain the model convergence, where i and j denote i-th and j-th medication in each visit, respectively.

Binary Cross Entropy Loss: The $\mathcal{L}_{bce}$ serves as the foundational loss function for optimizing CRHP's prediction accuracy:

$$\mathcal{L}_{bce} = -\frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{|M|} m_i^t \log(\hat{M}_i^t) + (1 - m_i^t)\log(1 - \hat{M}_i^t), \tag{23}$$

where $m^t \in \{0,1\}^{|M|}$ is the ground-truth set of drug recommendation, $\hat{y}^t$ denotes the prediction value of drug, T represents the visit number of a patient.

Multi-label Hinge Loss: Considering the potential correlation among medication labels, $\mathcal{L}_{mlh}$ is used to facilitate CRHP in learning the correct combinations of medication labels:

$$\mathcal{L}_{mlh} = \sum_{t=1}^{T}\sum_{M_i^t=1, M_j^t=1}\left[\frac{max(0, 1 - (\hat{M}_i^t - \hat{M}_j^t))}{|M^t|}\right]. \tag{24}$$

DDI Loss: The $\mathcal{L}_{ddi}$ is employed to guide the model in offering the safest possible medication recommendations:

$$\mathcal{L}_{ddi} = \sum_{t=1}^{N_m}\sum_{j=1}^{N_m}\sum_{i=1}^{N_m}(\hat{M}_i^t \cdot \hat{M}_j^t) \cdot A_e, \tag{25}$$

where A_e denotes the prior DDI adjacency matrix contained in the dataset.

Combining (23), (24), and (25), the joint loss function of the proposed CRHP can be denoted as follows:

$$\mathcal{L}_{joint} = \lambda_1 \cdot \mathcal{L}_{bce} + \lambda_2 \cdot \mathcal{L}_{mlh} + \lambda_3 \cdot \mathcal{L}_{ddi}, \tag{26}$$

where λ_1, λ_2, and λ_3 represent hyper-parameters controlling the weight of each loss function.

V. Experiments

A. Experimental Setup

1) Dataset and Preprocessing: The model's effectiveness is evaluated on two publicly available real-world datasets: MIMIC-III [37] and MIMIC-IV [38]. The MIMIC-III dataset collects data from adult patients admitted to the intensive care unit between 2001 and 2012, as well as some data from neonatal intensive care patients. The MIMIC-IV dataset is an extension of MIMIC-III with updated data and improvements to certain tables, which includes inpatient data from over 190,000 patients admitted to Beth Israel Deaconess Medical Center (BIDMC) between 2008 and 2019. These datasets include detailed information on diagnoses, procedures, medications, laboratory tests (e.g., blood glucose, creatinine), vital signs (e.g., heart rate, blood pressure), and clinical notes (e.g., nursing notes, discharge summaries). The MIMIC datasets are critical for advancing research in clinical decision support, predictive modeling, and healthcare AI, which provide a comprehensive and longitudinal view of patient care, enabling the development of models that can generalize to real-world clinical settings.

TABLE I
THE STATISTICS OF DATASET

Item	MIMIC-III	MIMIC-IV
patients	6,350	90,36
visits	15,032	20,616
diagnoses	1,958	1,892
procedures	1,430	4,939
drugs	131	131
Max of visits	29	28
Avg of visits	2.37	2.38
Avg of diag	13.63	17.37
Avg of proc	4.53	4.02
Avg of drugs	19.19	17.64

The original data of medication recommendation task is from "DIAGNOSES_ICD.csv", "PROCEDURES_ICD.csv", and "PRESCRIPTIONS.csv" files in the MIMIC datasets. To prepare the data for modeling, CRHP follows previous studies [17], [18] and performs extensive preprocessing. First, CRHP cleans the data by removing invalid entries (e.g., National Drug Code (NDC) of "0") and handling missing values using forward filling. Next, CRHP standardizes medication representations by mapping NDC to Anatomical Therapeutic Chemical (ATC) codes. CRHP also filters the data to retain only the most frequent diagnoses, procedures, and medications, enhancing training speed and facilitating easier analysis. For patients with multiple visits, CRHP ensures all records are preserved and sorted chronologically by visit time, using SUBJECT_ID and HADM_ID to uniquely identify each visit. This allows the model to capture longitudinal medical histories, which are crucial for modeling patient health trajectories and treatment outcomes. Finally, CRHP divides the MIMIC-III and MIMIC-IV datasets into three sets: training, validation, and testing sets, with proportions of 2/3, 1/6, and 1/6, respectively. The split is performed at the patient level to prevent data leakage, ensuring that all records of a single patient (including multiple visits, diagnoses, procedures, and medications) are contained within one set. Besides, CRHP obtains 40 of the most severe types of drug-drug interactions from TWOSIDES [39] to compute DDI score. This comprehensive preprocessing ensures data quality and consistency while leveraging the full context of patient histories for more accurate predictive modeling. The processed data statistics are outlined in Table I.

2) Baselines: For comprehensive evaluation of model efficacy, comparisons are made with the following baselines:

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- LEAP [10] integrates recurrent decoders and content-based attention mechanisms to secure multimorbidity drug recommendations.
- RETAIN [40] designs a two-level neural attention network to focus on the accuracy and interpretability of the model.
- DMNC [32] proposes a memory-augmented model to capture complex interactions of asynchronous sequential views for drug recommendation.
- GAMENet [17] introduces a memory module to integrate the DDI graph to enhance safe drug recommendations.
- SafeDrug [14] utilizes drug molecular architecture to model drug representations and designs a controllable DDI loss function to ensure the safety of drug recommendations.
- COGNet [18] designs a copy-or-predict mechanism to replicate historical drugs and generate accurate recommendations.
- MoleRec [25] models individual substructures' impact on patient representation to identify those substructures that are helpful in healing patients.
- DAI-Net [26] designs a dual adaptive interaction network that captures the interaction insights between patients' multifaceted health records and medication molecules.

3) Implementation Details: The proposed methodology is implemented utilizing PyTorch 1.9.0 on a Python 3.9.6 platform and evaluated on an Intel machine equipped with 1 NVIDIA 3080 GPU. Optimal hyperparameters are determined based on the validation set, specifically with a dimension size $dim = 64$. Model training uses the Adam optimizer [41] with a learning rate of 1×10^{-4}, a batch size of 16, and an iteration number of 50 epochs. In loss functions, the weights λ_1, λ_2, and λ_3 are set as 0.95, 0.05, and 0.2. To guarantee reproducibility, the random seed is set at 1203. Additionally, to ensure no overoptimization, the model employs 5-fold cross-validation and achieves consistent performance across all folds. Meanwhile, techniques such as early stopping and regularization are used to further mitigate overfitting and ensure model generalizability. During the testing phase, to ensure a fair comparison, CRHP employs bootstrapping sampling following [14], [18]. Specifically, CRHP randomly samples 80% of the test set data for each round of evaluation, and the results from 10 rounds are averaged to compute the mean and standard deviation, which are subsequently reported. To ensure the statistical significance of the experiment results, CRHP performs a paired sample t-test, and the obtained p-values are consistently below 0.05.

4) Evaluation Metrics: Drug recommendation performance is measured through four core metrics, with corresponding calculation formulas:

Precision and recall are utilized to compute the F1 score, which provides a harmonic balance between different types of prediction errors in the model:

$$Pre^t = \frac{|\{i|m_i^t = 1\} \cap \{j|\hat{o}_i^t = 1\}|}{|\{j|\hat{o}_i^t = 1\}|}, \tag{27}$$

$$Rec^t = \frac{|\{i|m_i^t = 1\} \cap \{j|\hat{o}_i^t = 1\}|}{|\{i|m_i^t = 1\}|}, \tag{28}$$

$$F1 = \frac{1}{N_k}\sum_{i}^{N}\frac{2 \times Pre^t \times Rec^t}{Pre^t + Rec^t}, \tag{29}$$

where $\hat{o}_i^t$ denotes the predicted drug label vector, m_i^t denotes the ground truth drug label vector, N_k denotes the number of patients.

The PRAUC is utilized to evaluate the model's performance in recognizing positive instances:

$$PRAUC = \frac{1}{N_k}\sum_{i}^{|M|}(Pre_{i+1}^t - Pre_i^t) * Rec_i^t, \tag{30}$$

where Pre_i^t and Rec_i^t denote the precision and recall rate with the prediction probability of the drug with the i ranking probability in the t-th visit as the threshold value.

The Jaccard metric is employed to consistently measure the similarity between the recommended drug set and the true drug set, aiming to validate the model's fitting ability at the set level:

$$Jaccard = \frac{1}{N_k}\sum_{t}^{N}\frac{|\{i|m_i^t = 1\} \cap \{j|\hat{o}_i^t = 1\}|}{|\{i|m_i^t = 1\} \cup \{j|\hat{o}_i^t = 1\}|}, \tag{31}$$

where the meanings of $\hat{o}_i^t$, m_i^t, and N are the same as those in the definition of F1.

The DDI rate represents the proportion of drug pairs with adverse interactions in the model's predicted results. In the adjacency matrix A_e, $(A_e)_{i,j} = 1$ indicates the presence of adverse interactions between the i-th drug and the j-th drug. A_e is used to calculate the DDI rate of the model's predicted results $\hat{o}_i^t$.

$$DDI = \frac{1}{N_k}\sum_{t}^{N}\frac{\{i, j|(A_e)_{i,j} = 1, \hat{o}_i^t = 1, \hat{o}_j^t = 1\}}{|M|^2}. \tag{32}$$

Additionally, the average number of medications (Avg medications) is reported to convey practical considerations [17], [25], hinting that the number of drugs cannot be infinitely increased to provide comprehensive coverage.

B. Main Experiments

Table II presents a comparative analysis of the performance of various methods on the MIMIC-III and MIMIC-IV datasets. Notably, CRHP surpasses all baselines in major metrics across both datasets, thereby underscoring the efficacy of the proposed model. Specifically, CRHP achieves relative improvements of 2.12, 1.91, and 1.79 percentage points in Jaccard, PRAUC, and F1-score on MIMIC-III, and 1.31, 1.83, and 0.98 percentage points on MIMIC-IV.

Specifically, the instance-based medication recommendation model LEAP exhibits performance limitations due to exclusive dependence on current medical information. In contrast, models like RETAIN, GAMENet, COGNet, and DGCL illustrate relatively superior performance by integrating longitudinal visit information in diverse ways. COGNet innovatively focuses on modeling patients' historical medication data, leading to improved recommendation performance.

Additionally, SafeDrug, MoleRec and DAI-Net incorporate external medical knowledge to enhance feature representation

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TABLE II
PERFORMANCE COMPARISON ON THE MIMIC-III/MIMIC-IV

Methods	Jaccard ↑	PRAUC ↑	F1-score ↑	DDI rate ↓	Avg medications
LEAP [10]	0.4521/ 0.4287	0.6549/ 0.5506	0.6138 / 0.5820	0.0731 / 0.0592	18.7138 / 11.5198
RETAIN [40]	0.4887 / 0.4239	0.7556 / 0.6798	0.6481 / 0.5791	0.0835 / 0.0939	20.4051 / 10.8602
GAMENet [17]	0.5237 / 0.4963	0.7775 / 0.7508	0.6783 / 0.6514	0.0861 / 0.0890	27.2145 / 18.4426
SafeDrug [14]	0.5233 / 0.5000	0.7742 / 0.7485	0.6764 / 0.6557	0.0615 / 0.0575	19.9178 / 14.4705
COGNet [18]	0.5289 / 0.5081	0.7675 / 0.7547	0.6829 / 0.6627	0.0831 / 0.0837	28.0900 / 24.6800
DGCL [13]	0.5255 / 0.4993	0.7738 / 0.7535	0.6801 / 0.6542	0.0836 / 0.0735	28.6253 / 16.6284
MoleRec [25]	0.5271 / 0.4930	0.7717 / 0.7426	0.6816 / 0.6503	0.0726 / 0.0961	21.6489 / 18.7085
DAI-Net [26]	0.5243 / 0.4955	0.7695 / 0.7419	0.6782 / 0.6532	0.0799 / 0.0765	21.8744 / 19.7963
CRHP	**0.5501* / 0.5212***	**0.7966* / 0.7730***	**0.7008* / 0.6725***	0.0633* / 0.0612*	23.4121 / 17.6654

↑ denotes the higher values are better, whereas ↓ represents the lower values are better. Numbers in bold and underlined indicate the current and previous best performance, respectively. * denotes that the result is statistically significant when compared with the second-best performance with p-value < 0.05.

TABLE III
ABLATION STUDY FOR DIFFERENT COMPONENTS OF CRHP ON MIMIC-III AND MIMIC-IV

	MIMIC-III			MIMIC-IV		
Methods	Jaccard	PRAUC	F1-score	Jaccard	PRAUC	F1-score
w/o CRAL	0.5394 ± 0.0012	0.7843 ± 0.0014	0.6887 ± 0.0012	0.5091 ± 0.0010	0.7615 ± 0.0013	0.6611 ± 0.0011
w/o HPI	0.4765 ± 0.0014	0.7348 ± 0.0013	0.6378 ± 0.0012	0.4563 ± 0.0009	0.6996 ± 0.0014	0.5988 ± 0.0012
w/o EDP	0.5385 ± 0.0011	0.7887 ± 0.0012	0.6924 ± 0.0012	0.5082 ± 0.0010	0.7593 ± 0.0012	0.6597 ± 0.0012
w/o MP	0.5428 ± 0.0011	0.7912 ± 0.0010	0.6961 ± 0.0013	0.5129 ± 0.0009	0.7670 ± 0.0011	0.6654 ± 0.0013
w/o DP	0.5438 ± 0.0012	0.7915 ± 0.0013	0.6959 ± 0.0010	0.5136 ± 0.0010	0.7675 ± 0.0012	0.6659 ± 0.0011
w/o HVM	0.5414 ± 0.0012	0.7900 ± 0.0011	0.6946 ± 0.0010	0.5109 ± 0.0011	0.7658 ± 0.0013	0.6639 ± 0.0010
CRHP	**0.5501 ± 0.0008**	**0.7966 ± 0.0010**	**0.7008 ± 0.0011**	**0.5212 ± 0.0008**	**0.7730 ± 0.0010**	**0.6725 ± 0.0011**

and improve recommendation performance. SafeDrug achieves lower DDI rates than most models on both datasets through integration of molecular drug knowledge and controlled DDI loss optimization. MoleRec, on the other hand, introduces a molecular substructure-aware encoding approach to enhance the security of the recommendations while maintaining accuracy. DAI-Net introduces patient's symptom data to learn coordinated associations between patient symptoms and medication molecules, improving drug representation. Nevertheless, simultaneously improving model accuracy while ensuring recommendation security remains challenging for above approaches. In contrast, the proposed CRHP achieves a lower DDI rate while also pursuing higher accuracy on both datasets, suggesting that the medications recommended by the proposed model are both safe and reliable.

C. Ablation Study

An ablation study is conducted to assess the effectiveness of the core components of the proposed CRHP model. The following variants are formulated:

- CRHP w/o CRAL, which excludes collaborative relation augmented learning and solely employs random initialization embedding for medical entity representation.
- CRHP w/o HPI, which excludes hierarchical prescription inference module to utilize current patient's condition to recommend drugs, neglects the modeling of patient's history information, and only use the stage-1 of entity-wise drug prediction.
- CRHP w/o EDP, which excludes the second stage of entity-wise drug prediction module in hierarchical prescription inference module.
- CRHP w/o DP, which excludes the disease perspective in hierarchical hierarchical prescription inference module.
- CRHP w/o MP, which excludes the medication perspective in hierarchical prescription inference module.
- CRHP w/o HVM, which excludes the history visit memory in hierarchical prescription inference module.

The ablation results, presented in Table III, reveal that each core component contributes to the CRHP's performance in drug recommendation. Notably, the performance of CRHP without CRAL decreases substantially, indicating that the collaborative relation augmented learning framwork is pivotal for improving recommendation accuracy. When the disease or drug perspective is removed from the HVM component, the model performance decreases, indicating that historical disease and drug representations contain valuable information for the current recommendation. Similarly, a significant performance drop is observed for CRHP without HPI, emphasizing the effectiveness of the model's disease and drug information fusion method in the recommendation process. In addition, when removing EDP and HVM separately, the model performance decreases, indicating that the feature representations captured from current and historical medical records are complementary and contain valuable information for final drug prediction. Overall, the complete CRHP model achieves the highest performance, demonstrating the effectiveness of the model's core components.

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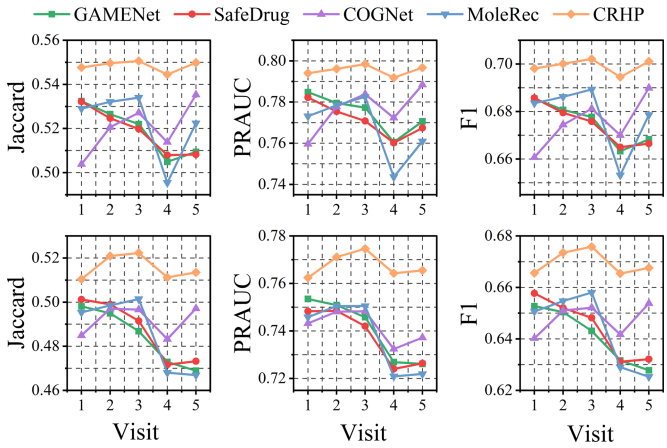


Fig. 2. The effect of history visit number for different models based on the MIMIC-III (first row) and MIMIC-IV (second row).

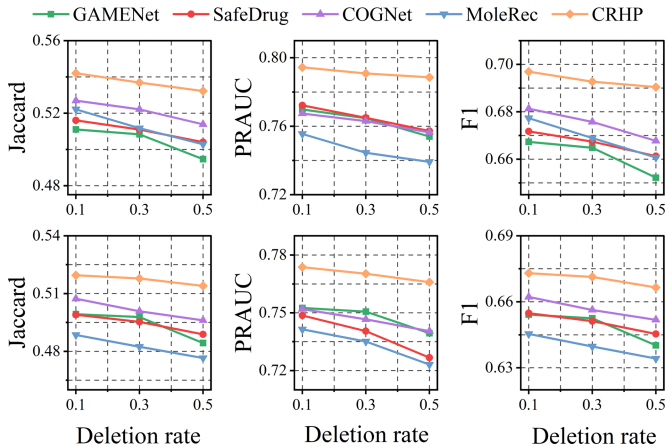


Fig. 3. Performance comparison of CRHP with different baselines for multiple EHR data deletion rates on the MIMIC-III (first row) and MIMIC-IV (second row).

D. Effect of Number of Visits

To evaluate the CRHP model's capacity for capturing historical medical information, the impact of visit number on model performance is examined. Experiments are conducted on two longitudinal patient datasets using four reproducible benchmarks (GAMENet, SafeDrug, COGNet, and MoleRec). Given that only a minute fraction of visits in the MIMIC dataset exceeds five, the analysis primarily focuses on integrating information from the first five visits.

As shown in Fig. 2, when faced with different numbers of visits in two datasets, the proposed model CRHP achieves optimal results compared to other models, demonstrating the capability of the model to extract relevant medical information from historical visits to enrich the current visit representation. When CRHP incorporates diverse longitudinal visit data, the overall performance surpasses that achieved when relying solely on information from the current visit on MIMIC-IV dataset. Notably, the incorporation of history records from the initial three visits results in the most significant improvement in model performance. Conversely, GAMENet and SafeDrug fall short in fully leveraging historical medication information, with the performance declining to varying degrees as the amount of historical information increases.

E. Robustness Analysis

To conduct the robustness of the CRHP model across diverse clinical settings, particularly those characterized by limited medical data and incomplete information, comparative experiments involving multiple models are performed on two datasets.

Given the stringent privacy and security concerns surrounding medical data, acquiring sufficient trainable data poses a significant challenge. To assess the model's performance and robustness in scenarios with scant training data, four prominent models (GAMENet, SafeDrug, COGNet, and MoleRec) are selected and conducted experiments on the MIMIC-III and MIMIC-IV datasets. The training data are systematically reduced by 10%, 30%, and 50%, with performance evaluated on the test set. Fig. 3 illustrates the performance trends of each model across different deletion rates on both MIMIC-III and MIMIC-IV. Notably, as the training data diminishes, all performance metrics decline. However, the model CRHP consistently outperforms the others when trained on the same proportion of data. Specifically, CRHP experiences minimal performance degradation compared to the other models under the same proportion of data reduction. In scenarios where the data undergoes various degrees of perturbation, the model CRHP demonstrates more excellent stability than the others on both datasets, affirming the robustness of CRHP.

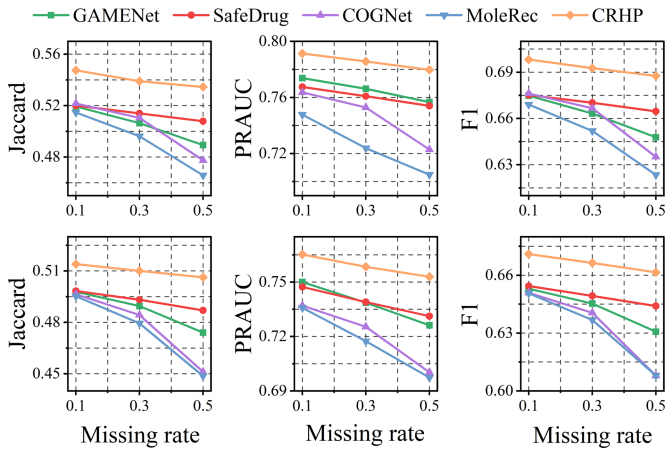


Fig. 4. Performance comparison of MRLN with different baselines for multiple EHR data missing rates on the MIMIC- III (first row) and MIMIC-IV (second row).

To further validate CRHP's capacity for handling missing electronic health records, experimental results are evaluated across various missing rates. Specifically, 10%, 30%, and 50% of the training set's diagnosis, procedure, and medication codes are masked to simulate patient EHR incompleteness, with performance assessed on the test set. Fig. 4 depicts the performance of each model across different masking rates. As the rate of missing EHRs increases, all performance metrics decrease. However, CRHP maintains the highest performance.

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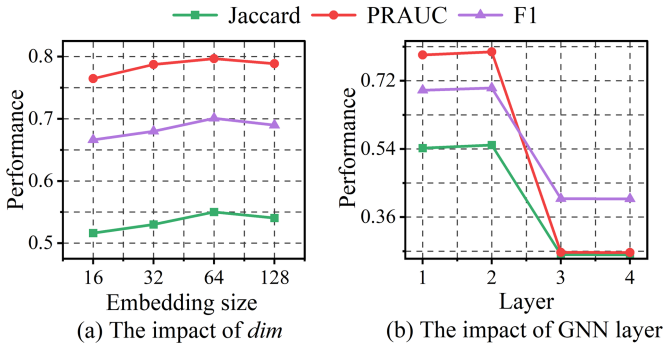


Fig. 5. Hyperparameter impact about different dimmension size (subfigure (a)) and GNN layers (subfigure (b)) for CRHP on MIMIC-III dataset.

Under identical experimental conditions, the proposed model exhibits greater stability than the others, indicating the model's robustness in handling missing EHR data. Among these models, GAMENet and COGNet experience significantly greater performance degradation as the proportion of missing labels increases. This may be attributed to these models recommending a larger number of drugs. When medical labels are disturbed, fully capturing the feature associations between disease labels and drugs becomes more challenging, and models that recommend more drugs are more susceptible to performance degradation compared to others.

F. Parameter Sensitivity Analysis

This subsection details the hyperparameter selection process for key model components, specifically examining embedding dimensions within [16, 32, 64, 128] and GNN layer numbers within $[1, 2, 3, 4]$. Model performance is evaluated using Jaccard, F1, and PRAUC metrics on the MIMIC-III dataset.

Fig. 5(a) presents the results of an extensive series of scaling experiments aimed at assessing the impact of the embedding dimension (dim) on the effectiveness of the proposed framework CRHP. The performance of all indicators first increases and then decreases with the increase of embedding dimensions and reaches the optimal level when $dim = 64$. Beyond this optimal point, however, an increase in embedding dimension leads to a decline in performance. The drop in performance may be traced back to two key factors. Firstly, as the embedding dimension increases, the model becomes more susceptible to noise, making the model more easily distracted by irrelevant information. Secondly, with higher dimensions, the model is also more likely to overfit. Consequently, the optimal embedding dimension for the framework is determined to be 64.

Fig. 5(b) illustrates a detailed analysis of the parameter sensitivity related to the number of GNN layers, with values ranging from 1 to 4. This analysis aims to determine the influence of GNN layer count on the performance of the proposed model. The results reveal a clear trend in performance as the number of GNN layers increases. Specifically, the model achieves optimal performance by configuring 2 GNN layers. When employing only a single GNN layer, the model's capacity to encapsulate intricate graph structures may be constrained, yielding suboptimal performance. Conversely, augmenting the number of GNN layers to three results in a rapid performance decrement, ascribed to phenomena such as over-smoothing and vanishing gradients, both pervasive issues in deep neural networks by excessive layering. Based on these comprehensive observations, the optimal configuration for the model consists of two GNN layers.

G. Case Study

Our case study analysis of a representative patient from the MIMIC-III dataset reveals several key advantages of CRHP compared to existing approaches. The patient's clinical course span two distinct visits: initially presenting with acute liver failure due to hepatitis B and cirrhosis, followed by a subsequent admission for chronic hepatitis B with atherosclerotic heart disease. This progression provides a good opportunity to evaluate how different models adapt to evolving clinical conditions. Diagnoses are represented using ICD-9 codes, while prescribed medications are denoted through ATC-third-level codes converted from drug NDC codes in the final recommendation. The "correct" denotes accurate prediction, "unseen" represents medications that are predicted but not present in the actual data, and "missed" signifies medications present in the actual data that are not predicted.

As demonstrated in Table IV, CRHP consistently outperforms baseline models (GAMENet, SafeDrug, and COGNet) in several clinically meaningful aspects. First, regarding accuracy, CRHP correctly identifies the highest number of essential medications in both visits, including crucial therapies like A06 A laxatives for liver failure management and C09 A ACE inhibitors for cardiovascular care. This performance advantage is particularly evident in the model's ability to maintain appropriate therapeutic coverage while avoiding excessive prescriptions. Second, CRHP demonstrates superior clinical relevance through its medication quantity predictions. Unlike GAMENet and COGNet, which tend to overprescribe (sometimes recommending nearly double the actual number of medications), CRHP maintains predictions much closer to real-world prescription patterns. This balance is clinically important as excessive polypharmacy can increase patient burden and safety risks. Conversely, while SafeDrug employs a controlled DDI loss mechanism to enhance safety, its overly conservative approach sometimes results in insufficient medication coverage that might compromise therapeutic efficacy.

The case study also reveals CRHP's nuanced understanding of disease-drug relationships. For instance, in managing the patient's liver conditions, CRHP appropriately includes hepatoprotective medications like A02B antacids. Furthermore, CRHP demonstrates clinically appropriate adaptation to the patient's evolving condition. When atherosclerotic heart disease is added to the diagnosis in the second visit, CRHP strategically modifies the medication regimen by: (1) maintaining essential cardiovascular medications including C01 C (cardiac stimulants) and C09 A (ACE inhibitors), which are important treatments for coronary artery disease; (2) appropriately continuing

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TABLE IV
AN EXAMPLE OF CASE STUDY

Diagnosis	Methods	Recommended Medication Combination
1st visit: Cirrhosis of the Liver, Acute Hepatitis B with Acute Liver Failure	Groundtruth(Num:16)	A01A, A06A, B05C, A12A, A12C, C01C, C03C, A12B, J01D, N07A, A11C, R06A, D07A, C03D, P01A, C02C
	GAMENet(Num:28)	9 correct + 7 missed + 19 unseen (N02B, A02B, A07A, N01A, C07A, N02A, J01M, B01A, C01B, N05C, D01A, B02B, N05A, A04A, M03A, R03A, N05B, R01A, J01C)
	SafeDrug(Num:9)	5 correct + 11 missed + 4 unseen (N02B, A02B, C07A, B01A)
	COGNet(Num:22)	10 correct + 6 missed + 12 unseen (N02B, A02B, A07A, M01A, N01A, C07A, N02A, A02A, B01A, C10A, C01B, A03F)
	CRHP(Num:15)	12 correct + 4 missed + 3 unseen (N02B, A02B, A07A)
2st visit: Chronic Carrier of Hepatitis B Virus, Atherosclerotic Heart Disease of Native Coronary Arteries	Groundtruth(Num:17)	A01A, A02B, A06A, B05C, A12C, A07A, J01D, A12A, C01C, M01A, N07A, A11C, C09A, R06A, V03A, S01E, M03B
	GAMENet(Num:10)	6 correct + 11 missed + 4 unseen (N02B, A12B, N02A, B01A)
	SafeDrug(Num:9)	5 correct + 12 missed + 4 unseen (N02B, C07A, A12B, B01A)
	COGNet(Num:10)	7 correct + 10 missed + 3 unseen (N02B, N02A, B01A)
	CRHP(Num:11)	10 correct + 7 missed + 1 unseen (N02B)

Green, red, and blue represent "correct," "missed," and "unseen," respectively.

hepatoprotective agents such as A02B (antacids/anti-ulcer drugs) given the persistent liver disease. This pharmacologically nuanced adjustment contrasts with baseline models: GAMENet inappropriately discontinues A12 C (mineral supplements), while SafeDrug fails to introduce C09 A (ACE inhibitors). COGNet, while maintaining some key medications, introduces non-essential drugs like C10 A (lipid modifiers) without clear clinical justification.

These observations collectively demonstrate that CRHP achieves an optimal balance between several clinically important factors: accuracy in identifying necessary medications, appropriateness in prescription quantity, and adaptability to changing clinical contexts. The model's performance in this case study suggests its strong potential for real-world clinical implementation, where such balanced decision-making is crucial for optimal patient care.

VI. CONCLUSION

This paper proposes a novel Collaborative Relation augmentation with Hierarchical Prescription inference network (CRHP) for effective medication recommendation. Specifically, CRHP first constructs multiple covariance knowledge graphs to capture fine-grained interaction relationships between different entities from a global perspective. Based on self-built knowledge graphs, CRHP designs a collaborative relation augmented learning module, which introduces hypergraph convolutional networks to capture high-order association information between different entities. Moreover, CRHP devises a hierarchical prescription inference module to synthesize current and historical clinical data through entity-level prediction and history visit memory for comprehensive medication recommendations. Extensive experiments on two publicly real-world EHR datasets, i.e., MIMIC-III and MIMIC-IV, show significant performance improvement brought by CRHP, which exceeds all baselines in major metrics. Furthermore, robustness studies and case analyses have also demonstrated the robustness and efficacy of the proposed model CRHP.

VII. LIMITATIONS AND FUTURE WORK

Although the proposed collaborative relation augmentation with hierarchical prescription inference network can improve medication recommendation models' performance, the current focus is incorporating intricate correlations between diseases and drugs to enhance feature representation. However, this approach neglects to explain how to perform prediction, limiting the model's interpretability. Also, the prevalent models for drug recommendation, which primarily employ deep learning methods, often suffer from limited interpretability, hindering clinicians' ability to understand the rationale behind the recommended medications.Clinical practitioners are more likely to trust model predictions when accompanied by medically relevant explanations. Additionally, the data is sourced from a single institution (BIDMC), which may limit the generalizability of the findings to other healthcare settings with different patient demographics or clinical practices. Meanwhile, certain rare conditions or treatments may be underrepresented in the dataset, potentially limiting the model's ability to accurately predict outcomes for these cases.

To address these limitations, future work will focus on: 1) Model Interpretability: Integrating explainable AI techniques, such as attention mechanisms or feature importance analysis, to enhance the transparency and trustworthiness of the proposed model for domain experts. 2) Multi-Center Validation: Collaborating with other institutions to validate the proposed model on diverse datasets, ensuring broader applicability. 3) Handling Rare Cases: Developing techniques, such as data augmentation or transfer learning, to improve model performance for underrepresented conditions. Addressing these challenges will enhance the interpretability, generalizability, and clinical relevance of the proposed framework.

REFERENCES

[1] H. Sun, S. Xie, S. Li, Y. Chen, J.-R. Wen, and R. Yan, "Debiased, longitudinal and coordinated drug recommendation through multi-visit clinic records," *in Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 27837–27849.

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[2] S. Bhoi, M. L. Lee, W. Hsu, H. S. A. Fang, and N. C. Tan, "Personalizing medication recommendation with a graph-based approach," *ACM Trans. Inf. Syst.*, vol. 40, no. 3, pp. 1–23, 2021.
[3] Y. Tan et al., "Metacare++ : Meta-learning with hierarchical subtyping for cold-start diagnosis prediction in healthcare data," in *Proc. 45th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2022, pp. 449–459.
[4] M. Xia, J. Wilson, B. Goldstein, and R. Henao, "Contrastive learning for clinical outcome prediction with partial data sources," *Proc. Mach. Learn. Res.*, vol. 235, 2024, Art. no. 54156.
[5] S. Wang, H. Lin, Y. Zhang, X. Li, and W. Qu, "MKFN: Multimodal knowledge fusion network for automatic ICD coding," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, 2023, pp. 2294–2297.
[6] X. Hou, Z. Liu, X. Li, X. Li, S. Sang, and Y. Zhang, "MKCL: Medical knowledge with contrastive learning model for radiology report generation," *J. Biomed. Informat.*, vol. 146, 2023, Art. no. 104496.
[7] X. Hou, X. Li, Z. Liu, S. Sang, M. Lu, and Y. Zhang, "Recalibrated cross-modal alignment network for radiology report generation with weakly supervised contrastive learning," *Expert Syst. with Appl.*, vol. 269, 2025, Art. no. 126394.
[8] Y. Zhang, X. Wu, Q. Fang, S. Qian, and C. Xu, "Knowledge-enhanced attributed multi-task learning for medicine recommendation," *ACM Trans. Inf. Syst.*, vol. 41, no. 1, pp. 1–24, 2023.
[9] Y. Tang et al., "LAMRec: Label-aware multi-view drug recommendation," in *Proc. 33rd ACM Int. Conf. Inf. Knowl. Manage.*, 2024, pp. 2230–2239.
[10] Y. Zhang, R. Chen, J. Tang, W. F. Stewart, and J. Sun, "LEAP: Learning to prescribe effective and safe treatment combinations for multimorbidity," in *Proc. 23rd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2017, pp. 1315–1324.
[11] F. Gong, M. Wang, H. Wang, S. Wang, and M. Liu, "SMR: Medical knowledge graph embedding for safe medicine recommendation," *Big Data Res.*, vol. 23, 2021, Art. no. 100174.
[12] X. Li, Y. Zhang, X. Hou, F. Meng, and H. Lin, "Multi-visit interactive re-calibration network for drug recommendation with a triple graph encoder," in *Proc. 2023 IEEE Int. Conf. Bioinf. Biomed.*, 2023, pp. 2040–2043.
[13] X. Li, Y. Zhang, X. Li, H. Wei, and M. Lu, "DGCL: Distance-wise and graph contrastive learning for medication recommendation," *J. Biomed. Informat.*, vol. 139, 2023, Art. no. 104301.
[14] C. Yang, C. Xiao, F. Ma, L. Glass, and J. Sun, "Safedrug: Dual molecular graph encoders for safe drug recommendations," in *Proc. Int. Joint Conf. Artifcial Intell.*, 2021, pp. 3735–3741.
[15] S. Bhoi, M. L. Lee, W. Hsu, and N. C. Tan, "Refine: A fine-grained medication recommendation system using deep learning and personalized drug interaction modeling," *Adv. Neural Inf. Process. Syst.*, vol. 36, pp. 24013–24024, 2023.
[16] Y. Tan et al., "4sdrug: Symptom-based set-to-set small and safe drug recommendation," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2022, pp. 3970–3980.
[17] J. Shang, C. Xiao, T. Ma, H. Li, and J. Sun, "Gamenet: Graph augmented memory networks for recommending medication combination," in *Proc. AAAI Conf. Artif. Intell.*, 2019, pp. 1126–1133.
[18] R. Wu, Z. Qiu, J. Jiang, G. Qi, and X. Wu, "Conditional generation net for medication recommendation," in *Proc. ACM Web Conf.*, 2022, pp. 935–945.
[19] L. Wang, W. Zhang, X. He, and H. Zha, "Personalized Prescription for Comorbidity," in *Database Systems for Advanced Applications*. Berlin, Germany: Springer, 2018, pp. 3–19.
[20] S. Wang, P. Ren, Z. Chen, Z. Ren, J. Ma, and M. de Rijke, "Order-free medicine combination prediction with graph convolutional reinforcement learning," in *Proc. 28th ACM Int. Conf. Inf. Knowl. Manage.*, 2019, pp. 1623–1632.
[21] Y. Ren, Y. Shi, K. Zhang, X. Wang, Z. Chen, and H. Li, "A drug recommendation model based on message propagation and DDI gating mechanism," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 7, pp. 3478–3485, Jul. 2022.
[22] F. Meng, X. Li, X. Hou, M. Lu, and Y. Zhang, "MGRN: Toward robust drug recommendation via multi-view gating retrieval network," *Bioinformatics*, vol. 40, no. 10, 2024, Art. no. btae572.
[23] T. Kim, J. Heo, H. Kim, K. Shin, and S.-W. Kim, "Vita:'carefully chosen and weighted less' is better in medication recommendation," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 8600–8607.
[24] X. Li, F. Meng, J. Wei, and Y. Zhang, "Towards robust drug recommendation based on dual perspective encoder and iterative denoising mechanism," *Expert Syst. with Appl.*, vol. 283, 2025, Art. no. 127784.
[25] N. Yang, K. Zeng, Q. Wu, and J. Yan, "MoleRec: Combinatorial drug recommendation with substructure-aware molecular representation learning," in *Proc. ACM Web Conf. 2023*, 2023, pp. 4075–4085.
[26] X. Zou, X. He, X. Zheng, W. Zhang, J. Chen, and C. Tang, "DAI-Net: Dual adaptive interaction network for coordinated medication recommendation," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 10, pp. 6201–6211, Oct. 2024.
[27] X. Li et al., "Knowledge enhanced representation learning network for drug recommendation," *Inf. Process. Manage.*, vol. 62, no. 5, 2025, Art. no. 104164.
[28] H. Zhang, X. Yang, L. Bai, and J. Liang, "Enhancing drug recommendations via heterogeneous graph representation learning in EHR networks," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 7, pp. 3024–3035, Jul. 2024.
[29] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Representations*, 2017.
[30] X. Zhang, B. Xu, F. Ma, C. Li, Y. Lin, and H. Lin, "Bi-preference learning heterogeneous hypergraph networks for session-based recommendation," *ACM Trans. Inf. Syst.*, vol. 42, no. 3, pp. 1–28, 2023.
[31] X. Zhang et al., "Price does matter! modeling price and interest preferences in session-based recommendation," in *Proc. 45th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2022, pp. 1684–1693.
[32] H. Le, T. Tran, and S. Venkatesh, "Dual memory neural computer for asynchronous two-view sequential learning," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2018, pp. 1637–1645.
[33] J. Shang, T. Ma, C. Xiao, and J. Sun, "Pre-training of graph augmented transformers for medication recommendation," in *Proc. Int. Joint Conf. Artif. Intell.*, 2019, pp. 5953–5959.
[34] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, and M. Wang, "LightGCN: Simplifying and powering graph convolution network for recommendation," in *Proc. 43rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2020, pp. 639–648.
[35] J. Jiang, Y. Wei, Y. Feng, J. Cao, and Y. Gao, "Dynamic hypergraph neural networks," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, 2019, pp. 2635–2641.
[36] X. Chen, K. Xiong, Y. Zhang, L. Xia, D. Yin, and J. X. Huang, "Neural feature-aware recommendation with signed hypergraph convolutional network," *ACM Trans. Inf. Syst.*, vol. 39, no. 1, pp. 1–22, 2020.
[37] A. E. Johnson et al., "MIMIC-III, a freely accessible critical care database," *Sci. Data*, vol. 3, no. 1, pp. 1–9, 2016.
[38] A. Johnson, L. Bulgarelli, T. Pollard, S. Horng, L. A. Celi, and R. Mark, "MIMIC-IV," PhysioNet, pp. 49–55, 2020, Accessed: Aug. 23, 2021. Online. Available: https://physionet.org/content/mimiciv/1.0/
[39] N. P. Tatonetti, P. P. Ye, R. Daneshjou, and R. B. Altman, "Data-driven prediction of drug effects and interactions," *Sci. Transl. Med.*, vol. 4, no. 125, pp. 125ra31–125ra31, 2012.
[40] E. Choi, M. T. Bahadori, J. Sun, J. Kulas, A. Schuetz, and W. Stewart, "Retain: An interpretable predictive model for healthcare using reverse time attention mechanism," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, vol. 29, pp. 3504–3512.
[41] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Representations*, 2015.

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